

Software Engineering and Data Science
SEIS 766: Vision AI
Assignment #2

Due Date: February 19th

Write a MatLab or Python program to answer the following questions **in sequence**:

1. Read in a gray-scale image from “HW_toyobjects_Gray.bmp” to a variable “**Img_RAW**”.
2. **Display** how many unique colors are in this gray-scale image “**Img_RAW**”.
3. Convert the gray-scale image “**Img_RAW**” into an index-color image with whatever number of colors you find out in the previous question.
4. **Display** how many unique colors are in your index-color image.
5. Resize the index-color image you obtained in question 3 to 180×180 and **display** how many unique colors are in this resized image.
6. Resize the index-color image you obtained in question 3 to 480×480 and **display** how many unique colors are in this resized image.
7. **Write** the index-color image you obtained in question 3 into a “HW_toyobjects_index.jpg” file.
8. **Write** the index-color image you obtained in question 3 into a “HW_toyobjects_index.bmp” file.
9. Read in an image from “HW_toyobjects_index.jpg” to a variable “**Img_JPG**”.
10. **Display** how many unique colors are in “**Img_JPG**”.
11. Read in an image from “HW_toyobjects_index.bmp” to a variable “**Img_BMP**”.
12. **Display** how many unique colors are in “**Img_BMP**”.
13. Show / **display** the three images in “**Img_RAW**”, “**Img_JPG**”, “**Img_BMP**”.
14. Make sure you can see the colors in those three images.

Submission Guideline:

1. Please include the answers to the above questions that required your program to “**display**” in a WORD document. Please also put **your name** on the top of your WORD document.
2. Please print your program (matlab or python) as **PDF / html** and include the **PDF / html** in your submission. Please name your program as “a2.m/.mlx/.py/.inpyb”.
3. Please also include your program in the formats like .m/.mlx/.py/.inpyb in your submission.
4. Prepare EVERYTHING mentioned in the guideline and submit them on **Canvas** no later than the due date. Please do **NOT** zip your files.
5. Please carefully follow the submission guideline. Otherwise, the instructor may not be able to grade your assignment.